HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Designed and implemented **Kubernetes controllers** and **operators** to automate the lifecycle management of platform services across hybrid cloud environments.
- Contributed to the development of a global workload placement system, enabling intelligent **scheduling** across **Kubernetes clusters** running in public cloud and on-premises data centers.
- Developed **Kubernetes APIs** and **Custom Resource Definitions (CRDs)** to provide clean, maintainable interfaces for platform capabilities, ensuring declarative system design.
- Built and maintained **reconciliation logic** within controllers to safely manage distributed system state, ensuring desired state convergence.
- Engineered **Python automation scripts** for data ingestion and scheduled reporting, processing approximately 1.5 TB of enterprise data weekly.
- Developed and maintained **Java Spring Boot REST APIs** for cross-service data exchange, incorporating request validation and circuit-breaker patterns.
- Containerized applications using **Docker** and deployed them to a managed **Kubernetes** cluster (EKS), configuring basic resource limits and health checks.

SKILLS

Cloud Native – Kubernetes, Docker, Helm, CRDs, Operators, Controller-runtime, Kubebuilder, API Extension

Languages – Go, Python, Java, SQL, R, Bash

Cloud Platforms – AWS (EC2, S3, Lambda, CloudWatch)

Databases – PostgreSQL, MySQL, Oracle

CI/CD & IaC – Jenkins, Git, Infrastructure as Code, GitOps

Observability – Splunk, Dynatrace, CloudWatch

Frameworks & Libraries – Spring Boot, pandas, NumPy, scikit-learn

PROJECTS

Kubernetes Operator for Service Lifecycle Management

- Developed a **Kubernetes operator** using Go and **controller-runtime** to automate the provisioning, configuration, and upgrading of a distributed messaging service.
- Implemented **CRDs** to define the desired state for the service, enabling declarative management of deployments across multiple clusters.
- Integrated **reconciliation logic** to ensure the operator maintained the target state, handling node failures and configuration drift automatically.

Automated Data Pipeline for Enterprise Systems

- Engineered a **Python ETL pipeline** to extract, transform, and load data from an enterprise ERP system into a PostgreSQL data warehouse.
- Developed **scheduled reporting scripts** using pandas to generate weekly operational summaries, reducing manual reporting effort by **~20 hours** per week.
- Deployed the pipeline components as **Docker containers** managed by Jenkins CI/CD, ensuring consistent execution and deployment.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)